

Question	Answer	Marks
4(a)(i)	temperature = $-273.15\text{ }^{\circ}\text{C}$	A1
4(a)(ii)	temperature = 0 K	A1
4(b)(i)	gas is ideal	B1
4(b)(ii)	$pV = NkT$	C1
	$N = 270 / (8.0 \times 10^{-21})$ $= 3.4 \times 10^{22}$	A1
4(b)(iii)	$n = (3.4 \times 10^{22}) / (6.02 \times 10^{23})$ $= 0.056\text{ mol}$	A1
4(c)	$\frac{1}{2} m\langle c^2 \rangle = (3 / 2) kT$	C1
	$\frac{1}{2} \times m \times 1900^2 = 1.5 \times 8.0 \times 10^{-21}$	C1
	$(m = 6.65 \times 10^{-27}\text{ kg})$	
	$m = (6.65 \times 10^{-27}) / (1.66 \times 10^{-27})$	C1
	$= 4.0\text{ u}$	A1

9702/41

Cambridge International AS & A Level – Mark Scheme

October/November 2025

PUBLISHED

Question	Answer	Marks